

Architecture Recap Short

Date: 2026-05-02

One-line summary

The current project implements a **shared spatiotemporal graph backbone** that predicts station-level temperature offsets and can optionally attach a fog / low-visibility head on top of the same latent state.

Core idea

The model is:

```
input meteorology -> temporal encoder -> spatial graph message passing -  
> shared latent state  
    -> offset head  
    -> optional fog head
```

This keeps **offset prediction as the backbone** while preparing the system for fog forecasting.

Inputs

Current default input is a 17-feature station-aligned meteorological vector:

```
mx2t, mn2t, era5_t2m, era5_d2m, UG_era5,  
dewpoint_spread_2m, rh_2m,  
era5_u10, era5_v10, wind_speed_10m,  
theta_v_2m, theta_v_delta_1d,  
t2m_delta_1d, dewpoint_spread_delta_1d,  
height, sin_doy, cos_doy
```

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The model accepts either:

- (N, F) single-step graph input
- (T, N, F) temporal graph input

Graph structure

- nodes = stations
- edges = directed k -nearest-neighbor connections
- edge features:

```
distance_km, delta_lat, delta_lon, delta_height
```

These edge features let the model use spatial distance and terrain gradients during message passing.

Backbone

The shared backbone has three stages:

1. Node encoder

- MLP projects raw features into hidden space

2. Temporal encoder

- Transformer self-attention over previous timesteps for each station
- learned temporal position embeddings
- attention pooling or last-step pooling

3. Spatial encoder

- stacked **TransformerConv** layers over the station graph
- residual connections and layer normalization

The result is one shared latent vector per station.

Outputs

Offset head

Primary implemented task:

```
[Delta Tmax, Delta Tmin]
```

Fog head

Optional downstream head:

```
fog logit or low-visibility logit
```

This is enabled only when fog labels are available.

Loss

Current training supports:

- masked MAE for temperature offsets
- optional BCE-with-logits loss for fog labels

Combined form:

$$L_{\text{total}} = L_{\text{offset}} + \lambda_{\text{fog}} * L_{\text{fog}}$$

Missing labels are masked, but stations remain in the graph.

Current status

What is already implemented:

- fog-relevant feature engineering
- temporal sequence support
- temporal self-attention
- spatial edge-aware message passing
- shared backbone with offset head
- optional fog head
- multitask loss path

What is still missing for a real fog paper:

- hourly fog / visibility labels
- full hourly merged dataset
- end-to-end fog training runs
- baseline comparisons

Correct description today

The project is best described as:

a fog-ready spatiotemporal graph backbone with offset prediction as the main implemented task
and fog prediction as the attached downstream head

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Recommended paper framing

Use this framing:

shared spatiotemporal backbone
-> offset correction head
-> fog / low-visibility head